



OPEN Explainable machine learning framework for assessing groundwater quality and trace element contamination in Eastern Saudi Arabia

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Groundwater quality in arid regions like Al Hassa, Saudi Arabia, is increasingly threatened by trace element contamination driven by human activity and natural geology. This study addresses the urgent need for data-driven tools to assess groundwater pollution in the region's multi-aquifer system. Groundwater samples were analyzed for trace elements and main physicochemical parameters. Using supervised machine learning (ML) models—Linear Regression (LR), Random Forest (RF), K-Nearest Neighbors (KNN), and Gradient Boosting Machine (GBM)—the Water Pollution Index (WPI) was predicted as a holistic metric of contamination. The GBM model outperformed all others, achieving a training coefficient of determination (DC) of 0.9970 and a mean absolute error (MAE) of 0.0017. During testing, it maintained a high DC of 0.9372 and an MAE of 0.0063, confirming its strong generalization ability. SHapley Additive exPlanations (SHAP) were used to rank feature importance and enhance model transparency. The most influential variables for WPI prediction were chromium (Cr, SHAP = 0.0214), aluminum (Al, SHAP = 0.0136), and strontium (Sr, SHAP = 0.0053), followed by Fe (0.0031), V (0.0028), and Se (0.0017). Despite generally acceptable water quality, elements such as Cr and Fe exceeded safe limits in several samples. This study presents a transparent, high-performing framework for groundwater quality assessment in arid conditions. The integration of explainable ML offers clear, actionable insights into sustainable water management and environmental decision-making.

Keywords Groundwater quality, Trace elements, Explainable machine learning, Water pollution index, SHAP analysis

Groundwater is the largest freshwater resource on earth and is essential for maintaining life. Groundwater, representing more than 97% of all freshwater, excluding glaciers and ice caps, is a critical reserve¹. However, the overexploitation of groundwater and other associated development activities have disturbed the interaction among groundwater, surface water, and seawater, which influences the overall hydrological balance, shaping both water quality and the health of ecosystems that rely on these interconnected resources. This hidden resource occurs in small underground spaces and fractures created either by the filtering down of water from the surface or entrapment through geological processes such as sedimentation or volcanic activity. Groundwater is the source of drinking water for half the global population and maintains ecosystems by providing water, nutrients, and a steadily stable temperature². It supplies much-needed water for food, energy, health, and recreation. Groundwater is used for irrigation in almost 100 million hectares of farmland, representing more than 40% of the water used in agriculture globally³.

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Human and environmental health are closely linked to the quality of available water resources⁴. The accumulation of transition elements, particularly heavy metals, in groundwater systems is a growing concern due to the harm they can cause to living organisms, including humans, animals, and plants⁵. These contaminants not only degrade water quality but also pose serious risks to public health and ecological stability^{6,7}. While certain trace elements like manganese, copper, zinc, nickel, and chromium are beneficial in small doses, they become harmful when present in quantities above accepted limits^{8,9}. Conversely, elements such as lead, cadmium, arsenic, and mercury are dangerous even at very low concentrations. Understanding the environmental pathways and behaviors of these elements—including how they move, persist, and impact health through the food chain—is crucial for managing their risks¹⁰. Their harmful effects are strongly influenced by factors like chemical form, structure, and how they bind in the environment, which affect their mobility and toxicity^{11,12}.

In recent years, trace elements have become a central focus in groundwater research, especially in areas where environmental conditions suggest elevated risks of contamination⁹. To control their impact, regular monitoring of their levels and distribution has become a necessary part of water resource management. This approach helps assess health risks and supports strategies for the sustainable use of groundwater. Various techniques have been applied to evaluate groundwater quality, including stoichiometric methods, statistical tools, index-based models, and chemometric techniques^{13–16}. However, conventional statistical methods may fail to capture the nonlinear relationships, multicollinearity, and high-dimensional interactions^{17,18}. More recently, advanced tools such as clustering methods, regression models, neural networks, and deep learning algorithms have been employed to predict and analyze water quality trends^{19–26}. Machine Learning (ML) techniques are particularly valuable because they can overcome these limitations by efficiently handling complex, nonlinear data, revealing hidden patterns, and providing higher predictive power compared to traditional methods^{27,28}.

In Saudi Arabia's Al-Hassa region, where groundwater is vital for domestic, industrial, and agricultural use, limited studies have addressed trace element pollution in its multi-aquifer system. Existing work has focused mainly on the area's hydrogeology, water salinity, and evaporation effects^{29–33}. Both human and natural factors drive groundwater contamination in this region. Industrial activities, agricultural inputs, and irrigation with treated wastewater contribute about 70% of the pollution, while the remaining 30% is linked to natural geological conditions and aquifer chemistry. This interaction between human actions and geogenic factors reflects the complex nature of groundwater pollution and highlights the need for targeted management responses³⁴. In this context, explainable artificial intelligence (XAI) becomes crucial. While ML models can provide high predictive accuracy, they are criticized as “black boxes,” offering little transparency into how predictions are made. SHapley Additive exPlanations (SHAP) address this limitation by quantifying the contribution of each feature and decision-making relevance^{35,36}. For groundwater quality studies, SHAP allows not only the prediction of contamination levels but also the identification of the most influential physicochemical parameters and trace elements that drive pollution trends.

This study presents a pioneering assessment of trace element pollution in the groundwater systems of the Al Hassa region, an area of immense ecological, agricultural, and socioeconomic importance in eastern Saudi Arabia. Groundwater in the region is essential as it serves as the primary source of water for domestic consumption, agriculture, and industrial activities, especially in the absence of reliable surface water supplies. Its importance is further heightened by the arid climate and limited rainfall, making groundwater a critical resource for sustaining livelihoods and economic development. Unlike previous studies that focused primarily on salinity and general hydrochemical attributes, this research provides a detailed and data-driven investigation into the behavior of trace elements within a complex multi-aquifer setting. This work delivers comprehensive groundwater quality dynamics by integrating the ML models and SHAP analysis. The introduction of SHAP-based explainable ML particularly allows for a transparent evaluation of feature contributions toward the WPI, offering not just predictions but meaningful interpretations. The significance of this study lies in bridging methodological innovation with environmental application, demonstrating how explainable ML can enhance groundwater management in arid regions. This framework bridges the gap between raw environmental data and actionable insights for water resource governance. The findings highlight the influence of anthropogenic pressures, demonstrating the urgent need for sustainable water management in the region. As water demand intensifies and contamination risks rise, the methodology and insights from this study offer a replicable and scalable model for other vulnerable groundwater-dependent areas across the Middle East and beyond.

Study area and data curation

Study area

The Al Hassa Oasis, located in the Eastern Province of Saudi Arabia, is a historically and ecologically significant area central to regional agriculture and water resources (Fig. 1). It lies approximately 320 km east of Riyadh and about 70 km west of the Arabian Gulf³⁷, with geographical coordinates between 49°30' to 49°50' E and 25°20' to 25°40' N. The terrain slopes gently from an elevation of around 130 to 160 m above sea level towards the coastal plain. Covering about 260 square kilometers, the oasis has a distinctive “L” shape—extending roughly 30 km from the south-northeast and 18 km from the west-east direction³⁸. The climate is classified as a subtropical desert climate, marked by extreme seasonal variations. Summers, spanning from May to August, are intensely hot with daytime temperatures reaching up to 45 °C, while winters, from November to March, are relatively cool with average lows of 8 °C. Annual rainfall is limited, averaging around 85 mm and occurring mainly during winter and spring. Humidity varies seasonally, with a yearly average of 38%. The oasis is supported by a multi-aquifer system that includes the Alat, Khobar, Umm Er Radhuma, and Neogene aquifers, which serve the needs of domestic, agricultural, and industrial use. Particularly, the Neogene aquifer is vital for the Al Hassa Irrigation and Drainage Project. Given the region's dependence on groundwater, managing water quality is essential. Therefore, in this study, the Al Hassa Oasis is selected as the focal point for modeling the WPI, considering its

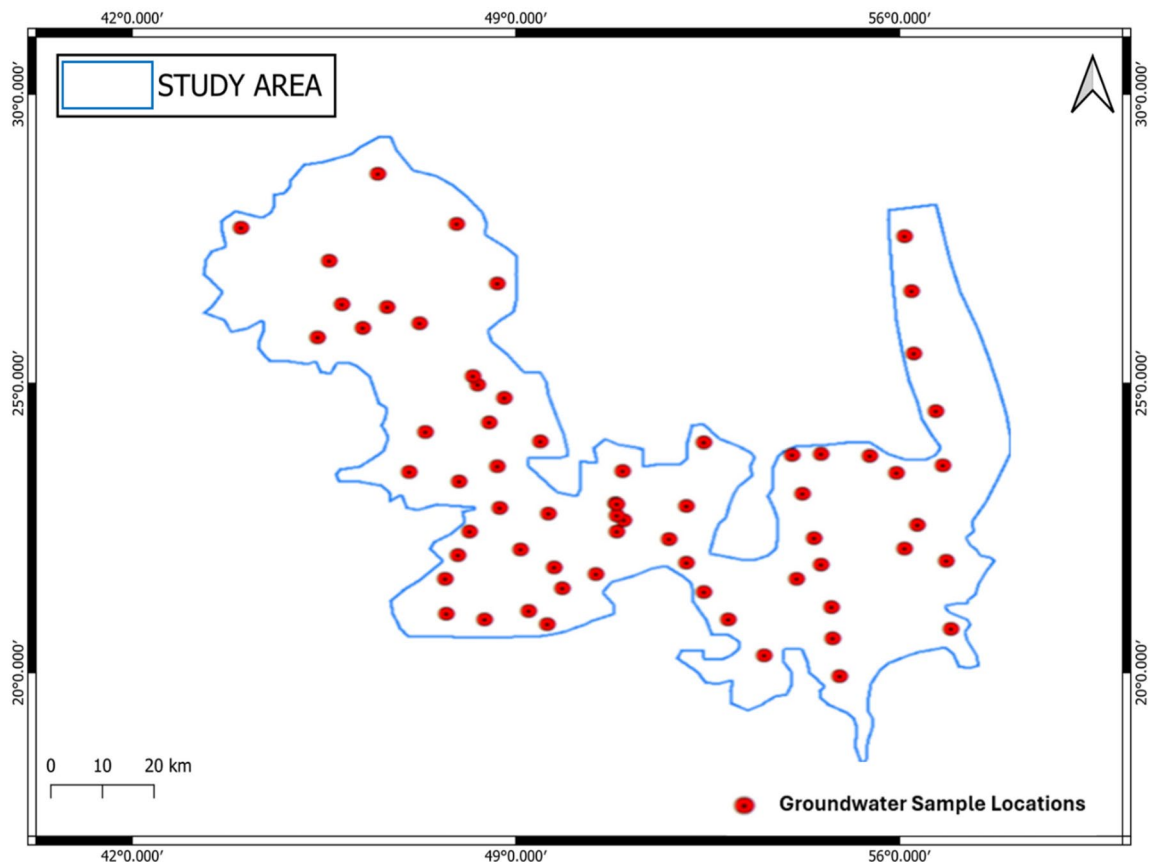


Fig. 1. Location of the study area and distribution of groundwater sampling. The boundary map data were obtained from DIVA-GIS (<https://www.diva-gis.org/>), and the map was prepared using QGIS software (<https://qgis.org/>).

vulnerability to environmental pressures and the strategic importance of groundwater in supporting agriculture and urban life.

Hydrogeology

The hydrogeology of Al Hassa is defined by layered sedimentary rocks that date back to the Late Cretaceous and Tertiary periods, including the Aruma, Rus, Neogene, Dammam, and Umm Er Radhuma formations. The Rus formation, made primarily of limestone, marl, and evaporites, separates the Umm Er Radhuma and Dammam formations by acting as a confining layer. These geologic units consist of carbonates like limestone and dolomite, along with anhydrite, gypsum, marl, and shale³⁹. The region contains four principal aquifers, each with different hydrogeologic characteristics. The Neogene aquifer system at the top combines less permeable clastic sediments and fractured, karstified rock. The underlying Umm Er Radhuma aquifer consists of well-developed karst features and fractured limestone, while the Dammam aquifer shows partial karst development. The Aruma aquifer also has fractured rocks but plays a more minor role³⁹. The Hofuf Formation within the Neogene unit varies in thickness between 20 and 100 m, whereas the Dam and Hadruk formations range from 100 to 200 m and are heavily fractured, contributing to groundwater storage through secondary porosity⁴⁰. Recharge happens mainly in the southwest, pushing groundwater northeast across the aquifers. These aquifers are the primary source of irrigation water for agriculture in the Al Hassa area, making their evaluation essential for understanding the pollution Index in the region.

Data curation and analysis

A groundwater sampling survey was carried out to evaluate the physical and chemical characteristics of water from the Neogene aquifer in the study area. On-site testing included key indicators such as pH, electrical conductivity (EC), total dissolved solids (TDS), temperature, and dissolved oxygen (DO), using a Hanna GPS Multiparameter Meter (H19829). To obtain accurate readings, each well was pumped for about 15 min before measurements began, ensuring that stagnant water was cleared. Groundwater samples were collected from different locations across the oasis, following random selection to represent diverse hydrogeological conditions. The samples were filtered in the field and stored in cooled containers at a temperature below 4 °C, in accordance with U.S. EPA recommendations for groundwater handling. In the lab, bicarbonate levels were analyzed using acid titration, while major and trace ions were determined through ion chromatography. The elements assessed included lithium (Li), boron (B), aluminum (Al), vanadium (V), chromium (Cr), iron (Fe), manganese (Mn),

nickel (Ni), cobalt (Co), copper (Cu), zinc (Zn), arsenic (As), selenium (Se), strontium (Sr), molybdenum (Mo), and barium (Ba). All measurements were expressed in milligrams per liter (mg/L). To derive a single quantitative measure of water quality, the WPI was computed using normalized parameter values and assigned weights as shown in Eq. 1 (13). The equation integrates the ratio of each parameter concentration to its permissible limit, multiplied by its relative weight, as shown below. This approach ensures comparability across different parameters and provides a robust index of cumulative groundwater contamination.

$$WPI = \sum_{i=1}^n \left(\frac{C_i}{S_i} \times W_i \right) \quad (1)$$

where C_i is the measured concentration of the parameter, S_i stands for the corresponding standard or permissible limit, and W_i is the weight assigned to the parameter.

The histogram plots in Fig. 2 illustrate the frequency distribution of physicochemical parameters and trace elements in the sampled groundwater. Most parameters show skewed or multimodal patterns. Table 1 indicates that pH values are stable, ranging narrowly between 7.03 and 7.22 (mean 7.07), suggesting neutral groundwater conditions across the study area. EC and TDS show wider variability, from 4359 to 7413 $\mu\text{S}/\text{cm}$ and 1979 to 4145 mg/L respectively, reflecting heterogeneous salinity linked to geology and land use. Among trace elements, Fe records the highest concentrations (1.93–2.87 mg/L, mean 2.55), followed by Sr (3.36–5.61 mg/L, mean 5.05), both pointing to strong geogenic influence. Cr concentrations vary between 0.026 and 0.057 mg/L (mean 0.049), exceeding permissible limits in several samples and marking it as a major contaminant of concern. Other elements such as Mn (0.0005–0.0032 mg/L), Co (0.0013–0.0023 mg/L), and As (0.0022–0.0052 mg/L) occur only in trace amounts with little dispersion. Overall, the statistics reveal stable acidity, heterogeneous salinity, and elevated levels of Fe and Cr as the primary drivers of groundwater quality variation in Al Hassa Oasis.

Materials and methods

Machine learning (ML) models

Linear regression (LR)

Linear Regression (LR) is one of the most basic algorithms used in ML for predicting continuous outcomes. It works by finding the best-fitting straight line through the data points, based on the relationship between input

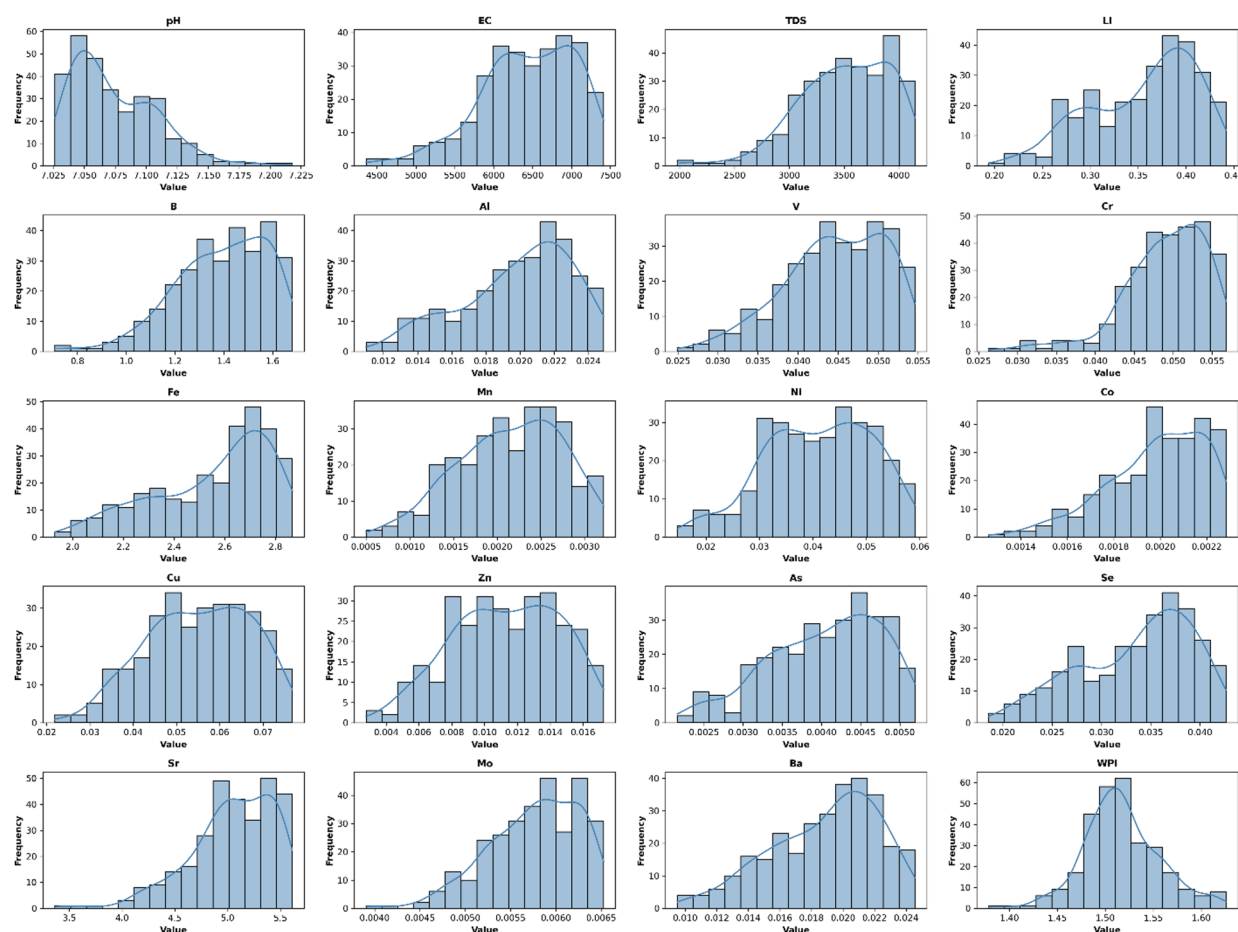


Fig. 2. Histogram distribution of groundwater quality parameters in Al Hassa Oasis.

	Mean	Median	Std Dev	Min	Max	Variance
pH	7.0742	7.0654	0.0345	7.0263	7.2180	0.0012
EC	6436	6506	614	4359	7413	376,844
TDS	3505	3533	416	1979	4145	172,729
Li	0.3556	0.3697	0.0552	0.1940	0.4419	0.0030
B	1.3909	1.4163	0.1902	0.7077	1.6808	0.0362
Al	0.0199	0.0205	0.0032	0.0110	0.0249	0.0000
V	0.0446	0.0450	0.0062	0.0249	0.0546	0.0000
Cr	0.0490	0.0498	0.0054	0.0263	0.0568	0.0000
Fe	2.5547	2.6263	0.2294	1.9301	2.8665	0.0526
Mn	0.0021	0.0022	0.0006	0.0005	0.0032	0.0000
Ni	0.0410	0.0417	0.0100	0.0146	0.0591	0.0001
Co	0.0020	0.0020	0.0002	0.0013	0.0023	0.0000
Cu	0.0552	0.0558	0.0119	0.0219	0.0768	0.0001
Zn	0.0113	0.0114	0.0033	0.0028	0.0173	0.0000
As	0.0040	0.0042	0.0007	0.0022	0.0052	0.0000
Se	0.0335	0.0348	0.0057	0.0186	0.0426	0.0000
Sr	5.0475	5.0984	0.3999	3.3632	5.6123	0.1599
Mo	0.0058	0.0058	0.0005	0.0039	0.0065	0.0000
Ba	0.0189	0.0195	0.0034	0.0095	0.0246	0.0000
WPI	1.5169	1.5133	0.0395	1.3784	1.6260	0.0016

Table 1. Descriptive statistics of physicochemical parameters and trace elements in groundwater samples.

variables and the target⁴¹. The method is used when the goal is to estimate the strength and direction of the effect of features on the outcome. It is also helpful for identifying possible trends or patterns in data over time. The model assumes a linear connection, where changes in the input features result in proportional changes in the output. It is mostly used because it is simple to comprehend and quick to apply, especially when the data does not show much complexity⁴². LR estimates the relationship between the dependent variable and independent variables using a linear equation of the form as shown in Eq. 2 (42).

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon \quad (2)$$

where Y is the target variable (WPI), X_1 are predictor variables, β_i are coefficients, and ϵ is the error term.

Random forest (RF)

Random Forest (RF) is a type of ensemble learning algorithm designed for both classification and regression challenges⁴³. In training, it grows a huge number of single decision trees and combines their predictions to achieve higher stability and accuracy of the output, as given by the Eq. 3⁴⁴. Its performance is based on two fundamental mechanisms: bootstrap aggregation (bagging) and random feature selection⁴⁵. Through these approaches, the model is made resistant to noisy input and irrelevant features, and thus does well on high-dimensional, complex datasets. RF has found extensive applications in a broad range of fields since it is not only very robust but also has high predictive performance even with unbalanced or partly missing data^{46,47}.

$$\hat{Y} = \frac{1}{T} \sum_{t=1}^T h_t(X) \quad (3)$$

where $h_t(X)$ is the prediction of the t^{th} decision tree, and T is the total number of trees.

Gradient boosting machine (GBM)

Gradient Boosting Machine (GBM) is a powerful predictive technique that builds an ensemble of decision trees in a sequential manner to enhance model accuracy⁴⁸. Unlike the parallel construction in RF, GBM trains each tree based on the errors of its predecessors, progressively reducing residuals with each iteration as shown in Eq. 4⁴⁹. This iterative approach allows the model better to learn intricate patterns and non-linear dependencies in the data⁵⁰. Its adaptability stems from the ability to fine-tune various hyperparameters, such as learning rate and tree depth, which significantly influence its predictive strength⁵¹. GBM is best known to exceptionally perform on structured problems of a dataset⁵².

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x) \quad (4)$$

where $F_{m-1}(x)$ is the previous model, $h_m(x)$ represents the new weak learner fitted to residuals, and γ_m is the learning rate.

K-Nearest neighbors (KNN)

K-Nearest Neighbors (KNN) is a non-parametric algorithm used for both classification and regression tasks. It makes predictions by comparing new data points to the most similar examples in the training set. The approach is based on the idea that similar inputs tend to produce identical outputs. It works well when the data is evenly scaled, and the input space is not too large. The number of neighbors (K) is a key setting that controls how many nearby points the model considers, as shown in Eq. 5⁵³. KNN does not build a model during training but instead stores all the data and makes decisions at the time of prediction. It is easy to implement and gives effective results when the data has clear patterns or clusters⁵⁴. The performance of KNN can depend on the choice of distance metric and the value of K. Table 2 provides the unique features of the ML models applied in this study, along with their specific relevance to groundwater quality prediction.

$$\hat{Y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y_i \tag{5}$$

where $N_k(x)$ is the set of k nearest neighbors of the query point x , and y_i are the observed values of those neighbors.

SHAP: SHapley additive exPlanations

SHAP (SHapley Additive exPlanations) is a powerful tool from the field of interpretable ML that quantifies how each input feature influences the output of a predictive model, as shown in Eq. 6⁶¹. Rather than providing a global importance ranking alone, SHAP attributes the output of individual predictions to the features involved, offering a more nuanced understanding of model behavior⁶². In this analysis, SHAP was employed to visualize and rank the contribution of features toward the WPI target, ensuring model transparency and insight into variable interactions. One of SHAP’s strengths lies in its ability to maintain consistency when interpreting how feature interactions influence predictions on a case-by-case basis⁶³. Unlike conventional importance scores, SHAP not only quantifies the magnitude of a feature’s effect but also clarifies its directional impact—whether positive or negative—on the predicted outcome.

$$\hat{y} = \phi_0 + \sum_{i=1}^M \phi_i \tag{6}$$

where ϕ_0 is the model’s baseline prediction and ϕ_i is the SHAP value for feature.

Performance metrics

To assess the prediction accuracy of the ML models for the WPI, six statistical metrics were employed: DC, MAPE, PBIAS, MSE, RMSE, and MAE. Each metric captures a different dimension of model performance, including error magnitude, consistency, and directional bias. The use of these metrics ensures a balanced and comprehensive comparison of the models across both training and testing phases.

Coefficient of Determination (DC)

$$DC = 1 - \frac{\sum_{i=1}^n (WPI_{obs,i} - WPI_{pred,i})^2}{\sum_{i=1}^n (WPI_{obs,i} - \overline{WPI_{obs}})^2} \tag{7}$$

Mean Absolute Percentage Error (MAPE)

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{WPI_{obs,i} - WPI_{pred,i}}{WPI_{obs,i}} \right| \tag{8}$$

Percent Bias (PBIAS)

Model	Unique Features	Relevance to Study
Linear Regression (LR)	Simple, interpretable model assuming linear relationships between predictors and response. Provides coefficients showing direction and magnitude of effects ⁵⁵ .	Serves as a baseline to compare nonlinear models, allowing assessment of whether nonlinear methods significantly improve predictions.
Random Forest (RF)	Ensemble of decision trees using bootstrap aggregation (bagging) and random feature selection; reduces overfitting and handles high-dimensional data ⁵⁶ .	Captures nonlinear and interaction effects among water quality parameters; robust against noise and missing data.
Gradient Boosting Machine (GBM)	Sequentially builds trees where each new tree corrects the residuals of the previous one; highly flexible with tunable hyperparameters (learning rate, tree depth) ^{57,58} .	Provides high predictive accuracy and captures subtle nonlinear relationships in groundwater datasets.
K-Nearest Neighbors (KNN)	Instance-based learner; prediction is based on similarity (distance metrics) between new data and training samples ^{59,60} .	Useful for capturing localized patterns in water quality data where similar conditions yield similar contamination levels.

Table 2. Unique characteristics and study relevance of the ML models.

$$PBIAS = 100 \times \frac{\sum_{i=1}^n (WPI_{pred,i} - WPI_{obs,i})}{\sum_{i=1}^n WPI_{obs,i}} \tag{9}$$

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (WPI_{obs,i} - WPI_{pred,i})^2 \tag{10}$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (WPI_{obs,i} - WPI_{pred,i})^2} \tag{11}$$

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |WPI_{obs,i} - WPI_{pred,i}| \tag{12}$$

Results and discussion
Evaluation of WPI prediction performance

The dataset was partitioned using a 70 to 30 ratio for training and testing, respectively. These partitioning also aimed at assessing both the learning potential and generalization ability of each model. Four supervised ML models were used in this study: LR, RF, GBM, and KNN. Predictive ability of each model was evaluated using six statistics: DC, MAPE, PBIAS, MSE, RMSE, and MAE. The DC ranges from 0 to 1, where DC = 1 indicates perfect agreement between observed and predicted values, and DC = 0 reflects no predictive power. Error-based metrics, such as MSE, RMSE, and MAE, measure the magnitude of deviation, with smaller values indicating higher accuracy. MAPE expresses prediction errors as percentages, while PBIAS reveals systematic overestimation or underestimation. Collectively, these metrics provide a comprehensive view of model accuracy, stability, and bias.

During the training phase, the GBM demonstrated superior performance across all evaluation metrics. It achieved the highest DC value of 0.9970, indicating its near-perfect ability to capture the relationship between the predictor variables and the target feature (WPI) (Table 3). Also, the model demonstrated the lowest MAPE of 0.1100, indicating a very small average percentage error compared to actual WPI values. The GBM further recorded a minimal MAE of 0.0017 and RMSE of 0.0022, reflecting high precision and low variability in its predictions. These performance characteristics are due to GBM’s capability to sequentially reduce bias and error through boosting weak learners. Additionally, it can handle complex interactions among features without requiring extensive preprocessing or feature scaling^{57,64}. LR and RF also performed well during training, achieving high DC values of 0.9795 and 0.9848, respectively, though with slightly higher error metrics than GBM. In contrast, KNN showed a noticeable lag, with a DC of only 0.4809 and MAPE exceeding 1.3, which points to poor learning capacity and model fit during training. This model also had the highest MAE and RMSE values, indicating larger average prediction errors and higher variability.

In the testing phase, the GBM was once again the best-performing model. It also maintained a high DC of 0.9372 and obtained the smallest RMSE of 0.0093, providing more validation on its great generalization ability to unexpected data. The smallest MAE of 0.0063 and MSE of 0.0001 were also obtained, providing additional support for the reliability of its practical application. These results demonstrate GBM’s resistance to overfitting and its ability to fit complex underlying patterns of the input features accurately. Such a characteristic renders GBM a suitable predictor of environmental indices with nonlinear relationships and subtle feature interactions. On the contrary, KNN was observed to yield the smallest DC of 0.2221 and the largest error among all measures of MAPE, RMSE, and MAE. These outcomes indicate that KNN was unable to establish valuable relationships

Model	DC	MAPE	PBIAS	MSE	RMSE	MAE
Training						
LR	0.9795	0.3074	0.0000	0.0000	0.0058	0.0047
RF	0.9848	0.1889	0.0072	0.0000	0.0050	0.0029
GBM	0.9970	0.1100	0.0000	0.0000	0.0022	0.0017
KNN	0.4809	1.3251	−0.0066	0.0008	0.0290	0.0203
Testing						
LR	0.9784	0.2904	0.0153	0.0000	0.0055	0.0044
RF	0.8993	0.5127	0.0100	0.0001	0.0118	0.0078
GBM	0.9372	0.4159	0.0226	0.0001	0.0093	0.0063
KNN	0.2221	1.5983	−0.0966	0.0011	0.0328	0.0243

Table 3. Comparative evaluation of ML models for predicting WPI.

within the training dataset and performed poorly in prediction. Such inadequacy is a direct result of the algorithm's sensitivity to local minima and lack of robustness when approaching high-dimensional and complex datasets. While KNN is often simple to apply and interpret, the algorithm does not favorably scale to problems that require modeling complex interactions or associations among variables. In comparison, the GBM provides a holistic and data-driven framework that strikes a balance between bias and variance, making it the most viable prospect for additional interpretation and feature exploration.

A radial plot, also referred to as a spider or radar chart, is a circular visualization that allows comparison of multiple quantitative variables across different categories⁶⁵. In this context, the radial plot, as shown in Fig. 3, is used to evaluate the performance of the four ML models across the standardized performance metrics. The plot presents the normalized values for both training (top) and testing (bottom) phases, allowing for a clear visual assessment of consistency and prediction quality. In these plots, MAPE, PBIAS, MSE, RMSE, and MAE are inverted during normalization so that higher values always indicate better performance for interpretability. The main trend observed is that GBM consistently maintains the most compact and centered shape in both plots. This indicates balanced and strong performance across all metrics. LR and RF also show competitive results, especially in the training phase, though they slightly diverge during testing. On the other hand, the KNN model clearly forms the widest and most irregular shape, particularly in the testing phase, indicating substantial performance deficiencies. Its pronounced extensions on the MAPE, RMSE, and MAE axes highlight high prediction error and instability.

Error plots in Fig. 4 display the residual distributions obtained to estimate the WPI. The left panel shows training residuals, while the right panel reflects testing residuals, both plotted against the sample index. Each residual represents the difference between the observed and predicted WPI values. A compact residual distribution centered near zero generally suggests higher model accuracy and lower error variance. During the training phase, GBM, RF, and LR produce tightly grouped residual lines that fluctuate gently around the zero line, indicating minimal errors and consistent performance across the samples. Comparatively, KNN exhibits broader and more frequent spikes, with some outliers outside the range of 0.1 or -0.1, indicating a greater spread and learning instability. These tendencies are repeated in the testing phase, with GBM also displaying a tight band of residuals with small variance, also favoring its ability to generalize well to new observations. RF and LR also demonstrate stable error behavior with moderate deviations. In comparison, KNN continues to display wide fluctuations with multiple peaks and troughs, characteristic of declining predictive ability on unseen observations. Both phase results thus verify GBM's performance at extracting the structure of the WPI dataset with slight bias and error, relative to KNN's comparative failures at displaying stability and precision on alternate splits of the dataset.

Figure 5 shows scatter plots comparing actual and predicted values of the WPI for both training and testing phases. The red dashed line represents the ideal one-to-one reference where predicted values would match the actual values exactly. The closer the points cluster around this line, the more accurate and reliable the model is. GBM displays the most consistent alignment with the one-to-one line in both training and testing, indicating strong predictive precision and generalization. LR and RF also show good alignment with minimal spread, though with slightly more variation in the testing phase compared to training. KNN, on the other hand, reveals a wider dispersion of points in both training and testing, especially with under- and over-predictions deviating clearly from the reference line. This wider scatter suggests lower accuracy and less stability in learning the relationship between input features and the WPI. The trend across all subplots reinforces that GBM yields the most reliable estimates, followed closely by LR and RF, while KNN performs less consistently across both data splits.

SHAP analysis

SHAP analysis is a valuable tool for explaining the relative contributions of input variables to complex prediction models. In environmental and water quality studies, this approach is particularly advantageous for identifying and prioritizing the chemical constituents and physicochemical parameters that exert the greatest influence on the variations in water pollution levels^{66–68}. By providing each feature with a numerical value that reflects its contribution to predicting the WPI, SHAP produces interpretability and informs monitoring prioritization and intervention targets. In Table 4, the features appear listed according to mean absolute SHAP values, a value that reflects their broad contribution to the model output according to the dataset. Cr, Al, and Sr possessed the most significant SHAP values and therefore appear to have the largest influence on outcomes of WPI predictions. These features have been most commonly associated with industrial effluents, metal corrosion, and rock-water interactions that all have the potential to affect the quality of waters within groundwater systems negatively. These exceptionally strong effects of Cr and Al are a function of their central role within the chemistry of waters and their potential to deteriorate human and ecological health when occurring at higher levels^{69,70}.

Among the other features, Fe, V, B, and Se also appear at the top positions. Their contribution, though less than that of Cr and Al, is still significant, and they have a natural geological origin or anthropogenic sources including agricultural runoff or wastewater pollution. Mn, Mo, and Li also highlight the role of trace metals in affecting groundwater quality. In addition, inclusion of pH, EC, and TDS among the top features highlights the importance of general physicochemical features in pollution assessment^{71,72}. These parameters reflect the general mineral load, salinity, and buffering activity of waters, which serve as quality indicators of water stability and suitability²¹. At the lower end of the ranking, Zn, As, and Ni recorded relatively low contributions within the model. Yet, their inclusion is still remarkable, considering their known toxicity levels in drinking water. Overall, this ranking not only supports the robustness of the model but also provides practical insight into which contaminants deserve priority in monitoring and remediation efforts within water management programs.

Water pollution in the study area remains a pressing concern due to the area's reliance on groundwater for domestic, agricultural, and industrial uses. The region's aquifer systems, which include the Neogene, Dammam,

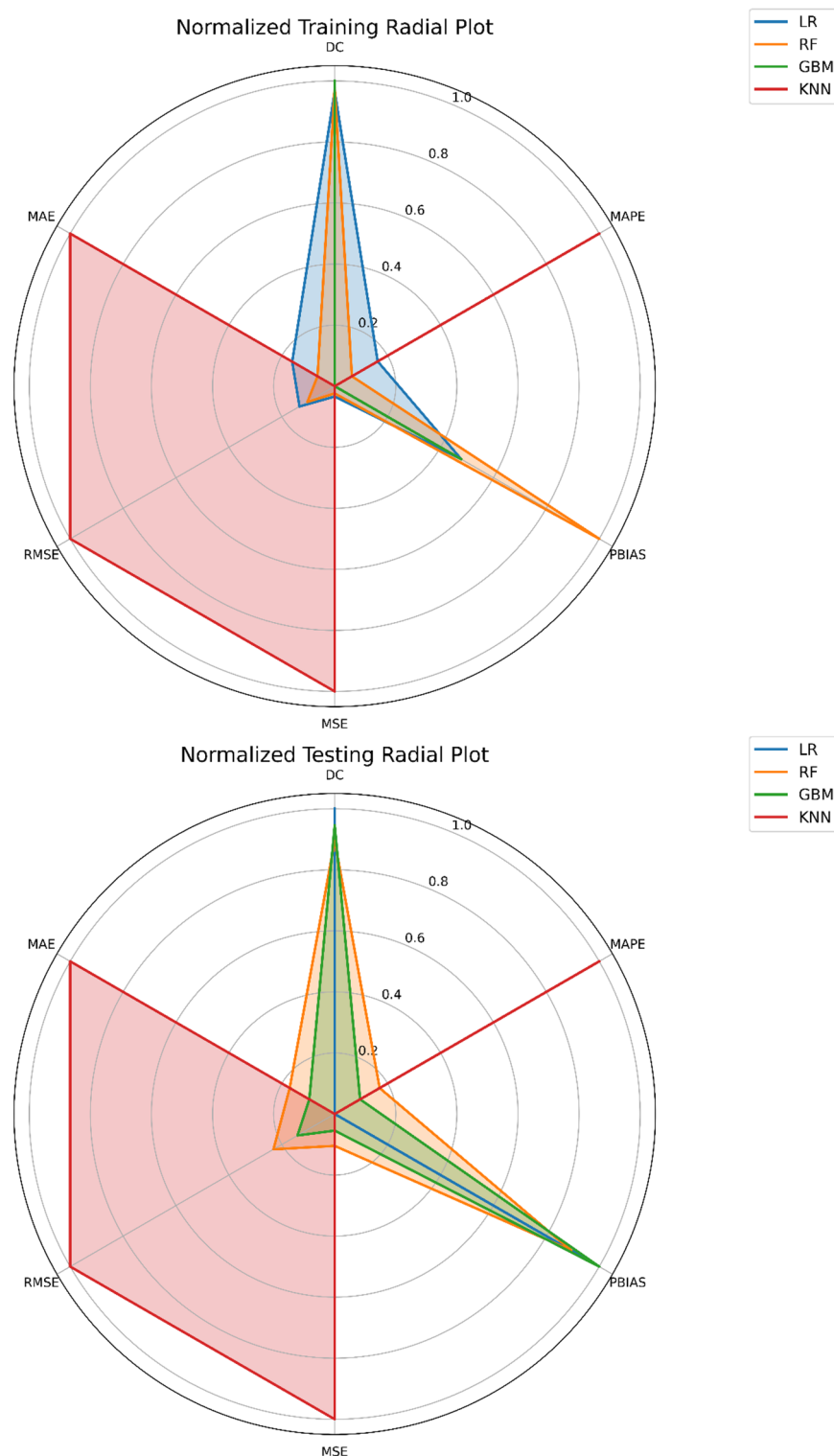


Fig. 3. Normalized radial plot showing the performance comparison of the ML models.

Rus, and Umm Er Radhuma formations, are geologically complex and vulnerable to contamination owing to their fractured and karstified nature. Findings show that although the overall groundwater quality is generally acceptable, some physicochemical parameters such as TDS, EC, and trace elements like Fe and Cr exceeded standard limits in several locations¹³. These elevated concentrations are attributed to both geogenic influences—such as mineral dissolution from the limestone-dominated aquifers—and anthropogenic sources, including agricultural runoff, wastewater reuse, and industrial discharges. Spatial analyses revealed that groundwater quality improves towards the northern part of the study area, while the eastern and southwestern zones are

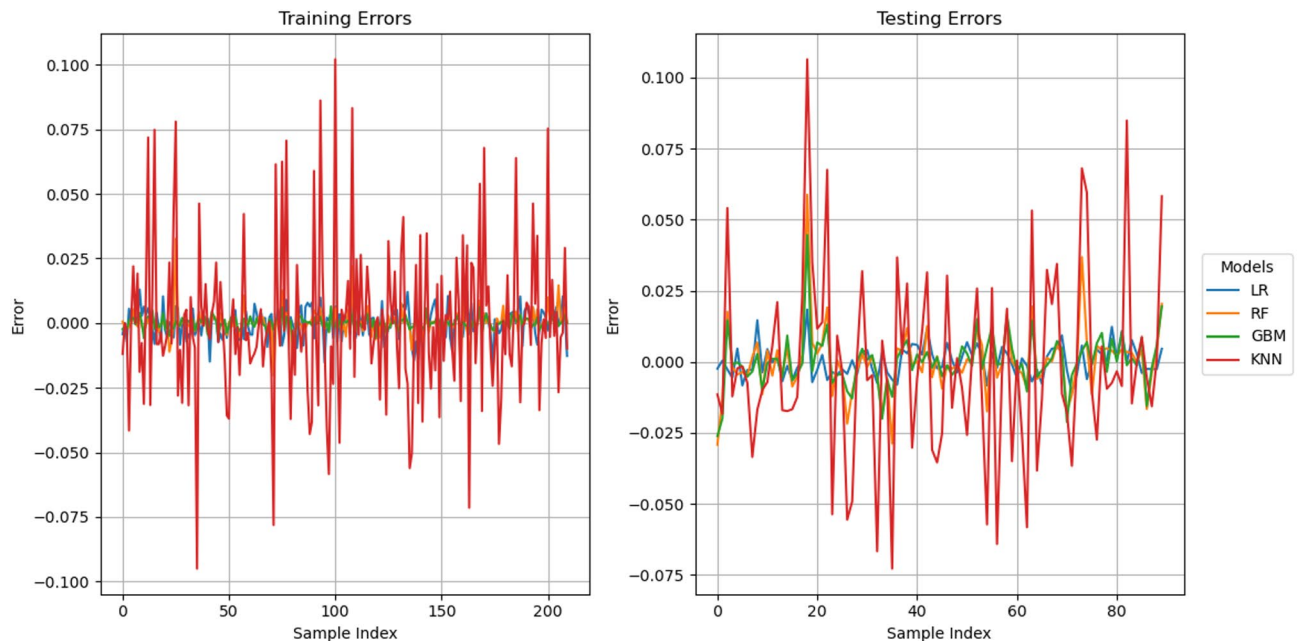


Fig. 4. Residual error distribution across training and testing phases.

more impacted by pollution, especially near urban centers⁷³. The presence of hazardous elements such as As, Ni, Co, and Se, even at trace levels, further underscores the need for regular monitoring due to their known health risks. The studies demonstrate that both natural processes and human activities contribute significantly to the observed pollution, and they emphasize the urgent need for groundwater protection strategies and sustainable management practices to safeguard this vital water source.

Figure 6 presents the SHAP beeswarm plot, which summarizes the distribution and direction of feature impacts on the predictive output of the model. Each dot represents a single observation, with its horizontal position showing the magnitude and direction of SHAP value, and its color indicating the relative feature value. Features are ordered by importance, with Cr, Al, and Sr having the highest overall influence on model predictions. Positive SHAP values suggest a tendency to increase the predicted WPI, while negative values suggest a decreasing effect. Cr shows the strongest positive impact when its concentration is high, which is expected due to its toxicity and common occurrence in industrial pollution. Al and Sr also demonstrate a similar trend, though with slightly less intensity. Fe, V, and B exhibit more balanced distributions, indicating both upward and downward influence depending on concentration. Se, Mn, and pH show moderate and concentrated effects, contributing to fine-tuning the prediction rather than driving it. The beeswarm plot reinforces that a few trace metals have dominant predictive roles, while others provide more supportive influence based on their interaction with local water chemistry.

Figure 7 illustrates a SHAP force plot, which explains the influence of individual features on a single prediction of the WPI. The horizontal axis represents the contribution of each variable toward increasing or decreasing the final WPI value. In this instance, the predicted value of the WPI is 1.50. Features pulling the prediction lower are shown in blue, while those with positive contributions are shown in red. The most significant contributors to lowering the WPI in this case are Sr = 4.83, Fe = 2.74, Li = 0.41, Se = 0.038, and Cr = 0.047. These elements pushed the prediction below the base value, indicating their influence in reducing the expected pollution level. The value of Al is 0.021 was the only feature that contributed positively, slightly increasing the final WPI value. The plot highlights how specific trace elements can impact the model's output and helps in understanding which elements play a key role in determining pollution severity at the local scale.

Conclusion and recommendations

Groundwater quality in the Al Hassa region of Saudi Arabia remains of critical importance for sustaining communities and agriculture in an arid environment with limited surface water resources. The application of explainable ML provided a robust framework for evaluating water quality, with the GBM demonstrating superior predictive performance (DC = 0.9970, MAE = 0.0017 in training; DC = 0.9372, MAE = 0.0063 in testing). SHAP analysis enhanced interpretability by identifying Cr, Al, and Sr as the most influential contributors to the Water Pollution Index, while also confirming the notable role of iron. Although groundwater quality was generally within acceptable limits, exceedances of Cr and Fe in several locations raise concerns regarding cumulative contamination risks. These outcomes highlight the combined influence of anthropogenic activities—particularly industrial discharge and wastewater reuse—on shaping groundwater quality. To safeguard this essential resource, regional authorities and water managers should adopt explainable ML tools for routine monitoring and prioritize high-risk elements such as Cr, Al, and Sr in remediation strategies. Expanding this framework to incorporate

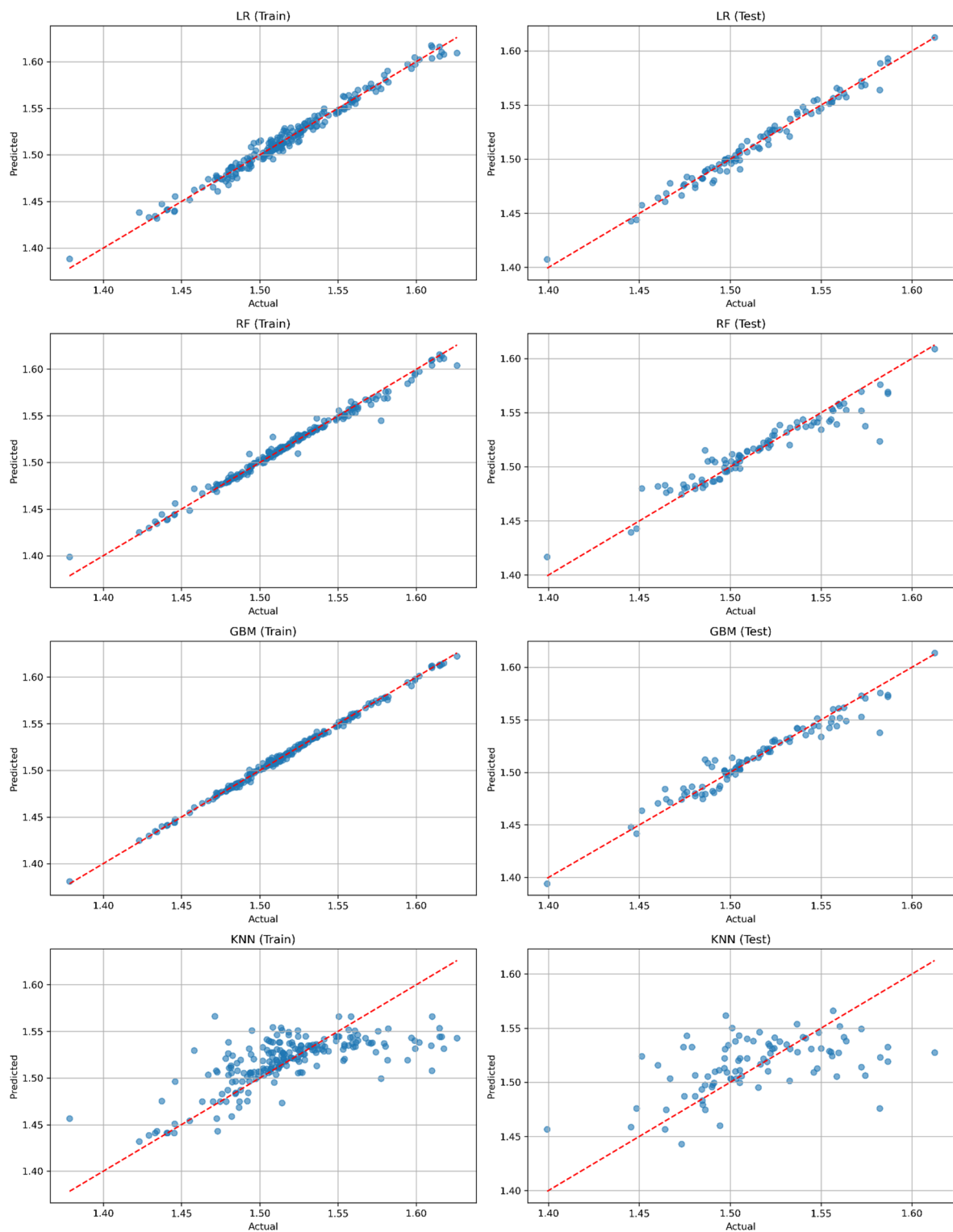


Fig. 5. Actual vs. predicted scatter distribution plots for training and testing sets.

the effects of climate variability and long-term environmental change will provide deeper insights into aquifer recharge, salinity evolution, and the resilience of groundwater systems under increasing stress.

Feature	Mean Absolute SHAP Values
Cr	0.021392
Al	0.013628
Sr	0.005321
Fe	0.003122
V	0.002797
B	0.001963
Se	0.001733
Mn	0.001517
pH	0.001434
Mo	0.001169
Co	0.001123
Li	0.001099
Cu	0.000905
Ba	0.000697
EC	0.000466
TDS	0.000298
Zn	0.000243
As	0.000223
Ni	0.000153

Table 4. Ranked feature importance based on mean absolute SHAP values for the WPI prediction.

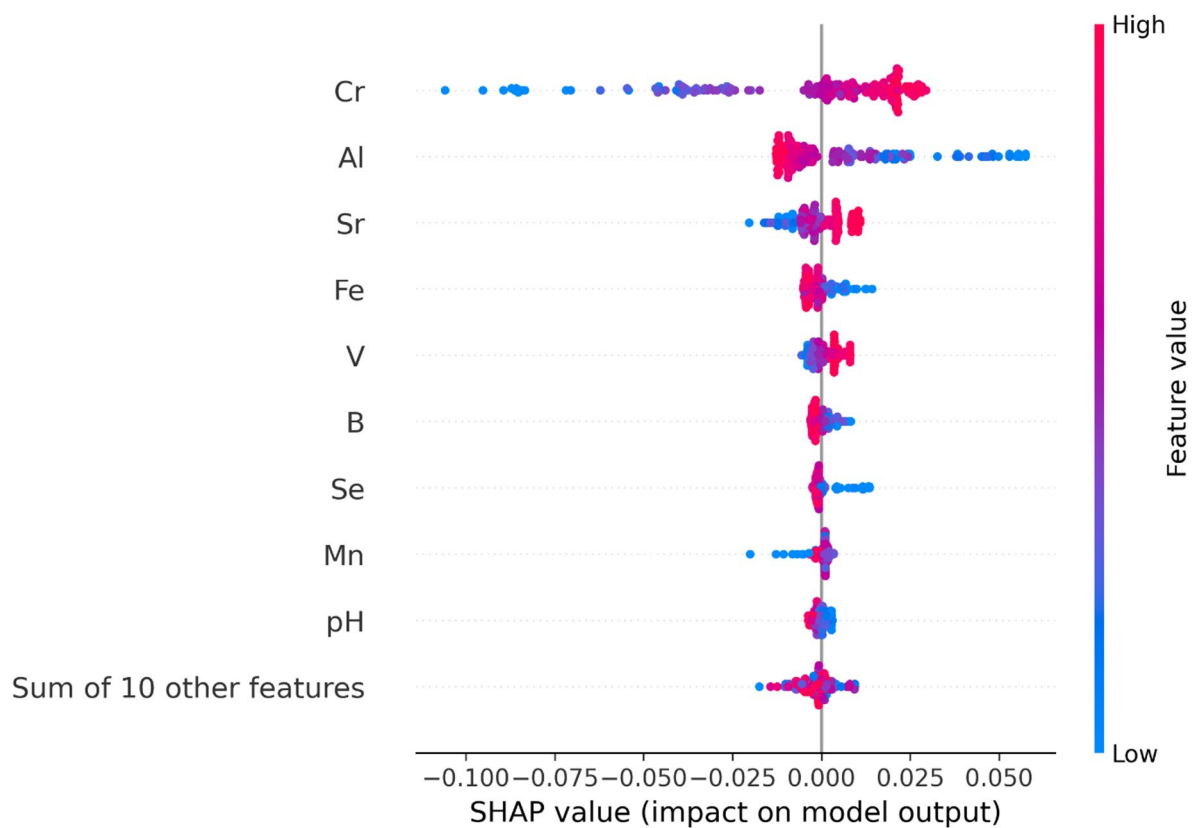


Fig. 6. SHAP Beeswarm plot showing feature impact distribution.

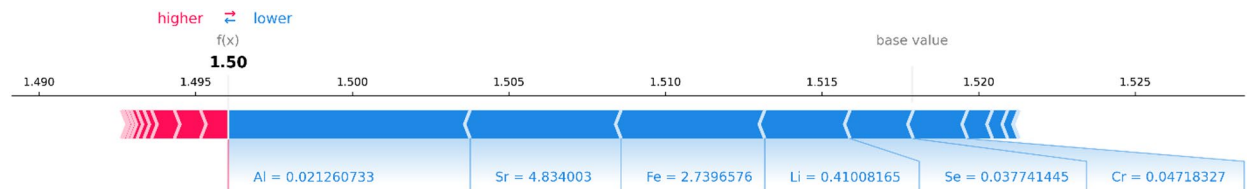


Fig. 7. SHAP force plot showing feature contributions to a single WPI prediction instance.

Data availability

The data is available upon request from the corresponding author (A.M. Jibrin, abdulhayatjm@gmail.com).

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Author contributions

A.A: Conceptualization, Methodology, Resource, Software, Writing- Reviewing and Editing. AM.J: Data curation, Software, Visualization, Validation, Writing- Original draft. S.D: Software, Validation, Formal Analysis, Writing- Reviewing and Editing. I.A.M: Methodology, Visualization, Validation. U.U.A: Data Curation, Writing- Reviewing and Editing, Resource. S.I.A: Data Curation, Formal analysis, Writing- Reviewing and Editing, Resource.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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